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Breakup of loosely bound nuclei at intermediate energies as indirect method in nuclear astrophysics: ^{8}B, ^{9}C and the S_{17}, S_{18} astrophysical factors

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We discuss the use of one-nucleon breakup reactions of loosely bound nuclei at intermediate energies as an indirect method in nuclear astrophysics. In particular, the breakup of ^{8}B and ^{9}C is described in detail in terms of an extended Glauber model. Existing experimental data for the breakup of ^{8}B at energies from 30 to 1000 MeV/u and of ^{9}C at 285 MeV/u on light through heavy targets are analyzed. The predictions of the model are in excellent agreement with independent reaction data (reaction cross sections and parallel momentum distributions for core-like fragments). Final state interaction has been included in the Coulomb dissociation component. Glauber model calculations using different effective interactions give consistent, though slightly different results. The results of this new analysis are compared with the earlier one for the astrophysical factor $S_{17}(0)$ for the key reaction for solar neutrino production ^{7}Be(p,γ)^{8}B. They are consistent with the values from most direct measurements and other indirect methods.

Radiative proton capture reactions are of crucial importance in nuclear astrophysics, and a large number of reaction chains were found to be needed in nucleosynthesis calculations for static [1] or explosive hydrogen burning scenarios [2]. This means that more data involving proton capture on unstable nuclei are necessary. In some cases direct experiments are possible, but in many more those are impossible with the present techniques and even with those of the foreseeable future. We have to rely on indirect methods instead. Recently we have proposed an indirect method to extract astrophysical S-factors from one-nucleon-removal (or breakup) reactions of loosely bound nuclei at intermediate energies [3,4]. It is based on the proof that the structure of halo nuclei is dominated by one or two nucleons orbiting a core [5,6]. Consequently, we use the fact that the breakup of halo or loosely bound nuclei is essentially a peripheral process, and therefore, the breakup cross-sections can give information about the wave function of the last nucleon at large distances from the core. More precisely, asymptotic normalization coefficients (ANCs) can be determined. Then, these ANCs are sufficient to determine the astrophysical S-factors for radiative proton capture reactions. The approach offers an alternative and complementary technique to extracting ANCs from transfer reactions [7], an alternative particularly well adapted to rare isotope beams produced using fragmentation. In this

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paper we discuss the use of existing experimental data on ^{8}B breakup at energies between 30 and 1000 MeV/nucleon [8–12] and on ^{9}C breakup at 285 MeV/nucleon [9], to determine the astrophysical S-factors S_{17} (which gives the reaction rate for the ^{7}Be(p,γ)^{8}B reaction of crucial importance for the solar neutrino question) and S_{18} (which gives the rate for the ^{8}B(p,γ)^{9}C reaction of importance for explosive hydrogen burning). The calculations presented in [3,4] on these two subjects have been extended and refined. First, the Coulomb part of the dissociation cross section was refined by including the final state interaction into calculations. Second, new data on the breakup of ^{8}B are analyzed [10–12]. Third and most important, a new set of calculations for the breakup of ^{8}B were made using five sets of different effective NN interactions. The new effective interactions considered, which do not involve any new parameters, give consistent results, but slightly different from one another. We consider these differences a good measure of the accuracy of the present (and other) indirect method(s).

In the breakup of loosely bound nuclei at intermediate energies, a nucleus $B = (Ap)$, where B is a bound state of the core A and the nucleon p, is produced by fragmentation from a primary beam, separated and then used to bombard a secondary target. In inclusive measurements, the core A is detected, measuring its parallel and transverse momenta and eventually the gamma-rays emitted from its deexcitation (see Fig. 1). Spectroscopic information can be extracted from these experiments such as the orbital momentum of the relative motion of the nucleon and the contribution of different core states, typically using Glauber models. The integrated cross sections have been used to extract absolute spectroscopic factors [6] or the ANC [3]. The latter approach has the advantage that it is independent of the geometry of the proton binding potential. The ANC C^B_{Ap} for the nuclear system $A + p \leftrightarrow B$ specifies the amplitude of the tail of the overlap function of the bound state B in the two-body channel $(A\,p)$ (see, for example [7] and references therein). Fortunately, this ANC is all we need to determine the astrophysical S-factor for the radiative proton capture reaction $A(p,\gamma)B$ which is a highly peripheral process. Using this approach we described the breakup of ^{8}B and ^{9}C in terms of an extended Glauber model. Some details for each case were published in [3,4]. The proton and the ^{7}Be (^{8}B) core are moving on a straight line trajectory and each interacts independently with the target. The breakup cross sections depend on the proton-target and core-target interactions and on the relative p-core motion. The wave function of the ground state of ^{8}B (^{9}C) is a mixture of $1p_{3/2}$ and $1p_{1/2}$ orbitals, around a ^{7}Be (^{8}B) core. The quantity $C^2_{tot} = C^2_{p_{3/2}} + C^2_{p_{1/2}}$ can be extracted from the breakup cross sections.

The ^{8}B ANC is extracted from existing breakup data at energies between 30-1000 MeV/u and on different targets ranging from C to Pb [8–12]. Two approaches were used. The first is a potential approach. To obtain the folded potentials needed in the S-matrix calculations we used the JLM effective nucleon-nucleon interaction [13]. We applied this technique for energies below 285 MeV/nucleon only and on all targets. In Figure 2 we show all ANCs extracted, consistent with a constant value. In a second approach, the Glauber model in the optical limit was used. The breakup process is treated as multiple elementary interactions between partners' nucleons, and the cross sections and the complex scattering amplitudes are taken from the literature. Calculations were done using different ranges for the elementary interactions: zero range, 1.5 fm ("standard"), 2.5 fm and individual ranges for each NN component ("Ray") [14]. No new parameters were adjusted. For details on the procedure see [4]. From the data at 936 MeV [12] we found the ratio $C^2_{exc}/C^2_{tot} = 0.16(4)$ which we used to assess the contribution of the ^{7}Be core excitation in all cases.

We found an average $C^2_{tot}(JLM) = 0.454 \pm 0.048$ fm^{-1} (fig. 2). Compared with Ref. [3] in this average we include the newer measurements by [10,11]. The value is in very good agreement with that determined before using the (^{7}Be,^{8}B) proton transfer reactions at 12 MeV/u [15]. The two values agree very well, in spite of the differences in the energy ranges and in the reaction mechanisms involved. The ANC extracted leads to the astrophysical factor $S_{17}(0) = 17.5 \pm 1.8$ eV· b for the key reaction for solar neutrino production ^{7}Be(p,γ)^{8}B. For the other effective NN interactions we checked that they correctly describe complementary data, like nucleon-target and core-target elastic and total reaction cross sections, where available. We rejected the calculations with zero-range and 2.5 fm range (they give too small or too large cross sections, respectively). For each of the two remaining NN interactions we find that all experiments give a consistent ANC, but the average values obtained are slightly different: $C^2_{tot}(\text{"}standard\text{"}) = 0.507$ fm^{-1} and $C^2_{tot}(\text{"}Ray\text{"}) = 0.517$ fm^{-1}. These differ by 11% and 13% respectively from the JLM value. We find no argument which value is best. If we take the unweighted average of all determinations we find an ANC that leads to $S_{17}(0) = 18.7 \pm 2.0$ eV· b. The uncertainties quoted are only the standard deviation of the individual values around the average.

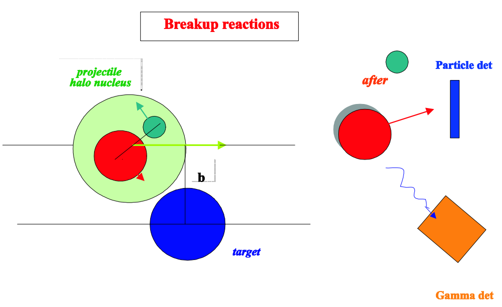

Figure 1. Schematic view of the breakup process and the geometry considered in the calculations

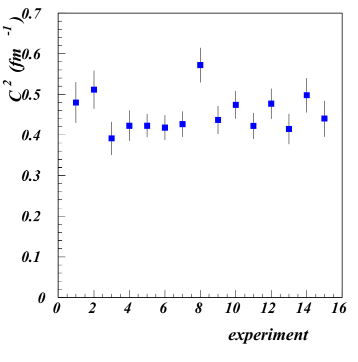

Figure 2. The ANCs determined from the breakup of ^{8}B at 28-285 MeV/u on C, Al, Sn and Pb targets, using the JLM effective interaction.

The same procedures have been applied for ^{9}C to determine the astrophysical S_{18} factor for the reaction ^{8}B(p,γ)^{9}C. The reaction is important in hot pp-chains as it can provide a starting point for an alternative path across the $A = 8$ mass gap [2]. The ANC for $^9\text{C} \rightarrow {}^8\text{B} + p$, has been determined using existing experimental data for the breakup of ^{9}C projectiles at 285 MeV/u on four different targets: C, Al, Sn and Pb

[9]. The introduction of the final state interaction in the Coulomb dissociation part does not change much the result, compared with the previous analysis [4]. We find now $C^2_{p_{3/2}} + C^2_{p_{1/2}} = 1.26 \pm 0.13$ fm^{-1}. To calculate the astrophysical S-factor we use the potential model. We find $S_{18}(0) = 46 \pm 6$ eV·b. A very weak dependence on energy is observed: $S(E) = 45.8 - 15.1E + 7.34E^2$ (E in MeV). This result is in very good agreement with other determinations [16,10], but not with one from Coulomb dissociation [17].

In conclusion, we show that breakup reactions at intermediate energies can be used to obtain astrophysical S-factors at stellar energies. There were many recent determinations of the key astrophysical factor S_{17}, but the precise value is still controversial. Our value is in agreement with those from all indirect methods and with most of the direct determinations (see the discussions in [18–20]), but one which stands out in its claim of a larger value and very small error [21]. Our results from the use of different NN interactions remind us of the fact that the precision of all indirect methods depends not only on the precision of the experiments but also on the accuracy of the calculations. Our findings may give a measure of the present status. However, we cannot find support for a value as large as 22 eVb.

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